Exploiting Visual-based Intent Classification for Diverse Social Image Retrieval

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ABSTRACT

In the 2017 MediaEval Retrieving Diverse Social Images task, we (TUD-MMC team) propose a novel method, named *intent-based* approach serving for social image search result diversification. The underlying assumption is that, each of the visual appearance of the social images is impacted by a photographic act, more concretely, why the image was taken. Better understanding the rationale behind photographic act could potentially benefit to social image search result diversification. To achieve this, we employed manual content analysis approach to create a taxonomy of intents. The experiment shows that a CNN-based neural networks classifier is able to capture the visual difference between the taxonomy of intents. We cluster images of Flickr baseline based on predicted intents, and aggregated search result by alternating images from different clusters. Our results reveal that, compared to conventional diversification strategies, intent-based search result diversification is able to bring a considerable improvement in terms of cluster recall with several extra benefits.

1 INTRODUCTION

The recent advance in deep learning, especially convolutional neural networks, has successfully been applied in various computer vision and multimedia tasks such as object recognition and scene labeling [4]. However, recognition of literally depicted content of multimedia documents (i.e., what is visible in the image) has absorbed most of the research attention. In contrast, less research has focused on social, affective and subjective properties of data, for example, why the image was taken.

In this paper, we introduce user intent, i.e., the goals that users are pursuing when they take photos, has visual reflexes that can be captured by automatic visual classifiers, and intent classes can be further applied to search result diversification [5] since the goals of the photographer provides a simple, easily understandable explanation for the differences observed between photos [8].

However, given the fact that the lack of intent taxonomies (definitions of intent classes) and data sets annotated with intent labels. We will start with creating a taxonomy of intent classes. The intent discovery process will be discussed in section 2.

2 INTENT DISCOVERY

2.1 Data Set Generation

The intent taxonomy was created using a manual content analysis [6] approach on the basis of YFCC100M [11], the largest public image collection that has ever been released. Since we are interested in build a taxonomy of intent classes with higher abstraction level that goes beyond concept detection, we choose to use NUS-WIDE concepts [2] that serve as queries to query images from the YFCC100M data set with a tag-based retrieval system.

For each retrieved documents associated with NUS-WIDE concepts list, we collect top-200 relevant images. We use the entire ranked list if less than 200 images can be found. As a result, for 81 NUS-WIDE concepts, we arrived at a data set of 15618 images.

2.2 Intent Labeling

The data set was produced by an expert annotator who examined the images in turn. For each image, we conduct manual content analysis and assign a preliminary intent label. When a new image comes in, it is then judged as either belonging to an existing intent class, or requires to create a new intent class. Before introducing a new class, the annotator returns to the previous annotated images to ensure that it is not possible to accommodate the new image by updating the description of an existing class. If no existing class can be extended to accommodate the new image, a new intent class is introduced. The final 14 classes intent taxonomy is shown in [12].

3 INTENT CLASSIFICATION

We adopt a conventional transfer learning scheme to predict the intent class of an image. Transfer learning trains models on one task, and leverages them for a different, but related task [7]. In our case, we used VGGNet [10] to extract visual content features from our images (originally trained on ImageNet [3]). The last fully connected layer (between 2048 neurons and 1000 class scores) was removed and the rest of the network serves as a feature extractor. We retrained a Softmax classifier using a cross-entropy Softmax loss on our image data set annotated with 14 intent classes using 70% of the intent data set. Meanwhile, 25% of the images were held out for validation purposes. (The remaining 5% are not used here.) Before we trained, we re-sized all images to 224x224 pixels, and applied data augmentation (random horizontal flipping, chopping and re-scaling). Our model achieved 71% accuracy on the validation set, suggesting that intent classes are visually stable enough to allow a classifier to generalize over them.

4 DIVERSIFICATION

The intent-based search result diversification works as follows: The first step is to create a refined initial ranked list by re-ranking the Flickr baseline using textual features (vector space model with tf-idf weights) with the aim of increasing precision. After that, the top N images are classified based on our intent classifier trained for

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Data Set	Evaluation	**visual**	**text-rerank + visual**	**text-rerank + text**	**text-rerank + intent**
Dev Set	Precision@20	61.52%	67.72%	67.72%	67.69%
	CR@20	49.29%	53.61%	52.36%	55.61%
	F1@20	54.73%	59.83%	59.05%	61.07%
Test Set	Precision@20	66.01%	70.71%	70.36%	72.62%
	CR@20	56.98%	58.09%	61.42%	61.25%
	F1@20	58.30%	61.21%	63.43%	64.62%

Table 1: Results in terms of Precision, Cluster Recall and F1 score with respect to 4 different runs on Dev and Test set.

predicting photographer's intent. Once the predictions are made, we cluster the first N results based on intent classes. Following this step, we pick the top-one photo without replacement for each cluster, under the assumption that new clusters reflect diversity as captured by photographer's intent. Finally, these new clusters are sorted internally based on the textual re-ranking position of the images and concatenated to create the final re-ranking list.

In addition to the intent-based approach, we also submitted three runs: *visual run*, *text-rerank + visual run* and *text-rerank + text run* for search result diversification.

Regard as *visual run*, we directly apply k-means clustering on CNN-based descriptor provided by task organizer. We employed a heuristic approach to initialize the number of k, that is, we treat k as a variable and initialize $k \in (1, n]$ and apply k-means clustering for n times. For each time, we evaluate clustering performance with silhouettes analysis [9] and select the best k with respect to silhouettes score.

Our *text-rerank+visual run* adopts the same general strategy of visual-based approach, the difference is that instead of directly apply k-means clustering, we re-ranked the Flickr baseline with tf-idf weights in ahead.

For our text-based approach, again, we re-ranked Flickr baseline with tf-idf weights. Since in this case, we are not allowed to use visual descriptors, the most critical issue is to learn a good representation for each "short document" consist of title, description and tags. To achieve this, we adopted the idea named *weighted word embedding aggregation* proposed by Cedric et al. [1]. More concretely, for each term associated with an image, we use its 50-dimensional word embedding vector. (Word embedding vectors were suppplied by the organizers.) Each image is thus represented as a set of vectors. For an image with m terms we have set of m 50-dimensional vectors. To model an image, we take the coordinate-wise maximum and minimum of the set of m vectors, and concatenate the resulting maximum and minimum vectors to arrive at a 100-dimensional vector, which is our final text-based image representation. For each query, we have a set of 300 image vectors, to which we apply k-means clustering with silhouette analysis.

5 RESULTS AND ANALYSIS

Table 1 reports the results in terms of the official MediaEval 2017 evaluation metric Precision@20, CR@20 and F1@20. In general, higher precision usually is associated with relatively higher cluster recall and F1 scores because non-relevant images have no associated diversity cluster label (i.e., d-gt). This phenomenon can be clearly obeserved comparing *visual* and *text-rerank+visual*. What is surprising is that the text-based image representation achieves a better clustering result on the test set compared to the visual CNN represention. The clustering result of *text-rerank+text* and our intent-based strategy achieves on the test set is approximately equal. The intent-based approach appears to give a boost to relevance as measured by Precision@20 and F@20.

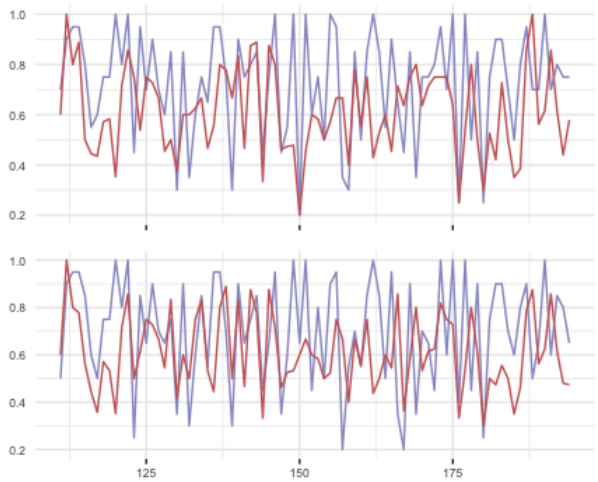

Figure 1: Comparison between *text-rerank + intent run* (up) and *text-rerank + text run* (bottom) over all query id (x-axis), purple line represents for P@20, red line represents for CR@20.

Figure 1 shows that both metrics fluctuate widely with respect to different queries. We measured the Pearson coefficient between P@20 and CR@20 for *text-rerank+intent* (0.41) and *text-rerank+text* (0.35), which reveals that the intent-based approach is more sensitive to initial ranking precision. The standard deviations are comparable: $\sigma = 0.17$ for *text-rerank+text* and $\sigma = 0.18$ for *text-rerank+intent*.

We point out three other aspects of the intent-based diversification approach that make it practically useful. First, intent-based diversification has the advantage of better understandability since the classification result is able to directly provide a user-interpretable indication of the reason behind the ranking. The retrieval system can provide the user with an explanation for its prioritization of search results. Second, once the model has been trained, we do not necessarily need to fine-tune the hyper parameters, i.e., the position to cut the dendrogram (for hierarchical clustering) or initial the number of k (for k-means clustering). Third, image labels are generated off-line at indexing time, and a clustering step at query time, which increases the system response time, is not necessary.

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